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Mobile Forensics (MBF 712S) Lab Investigation	
Lab No:	*Lab 2*
Lead Investigator:	Barnabas Iihuhwa 223081701 *Titus Shaangelo 222084774* *Tjijenda.u 220113009*
Qualifications:	Bachelor of Computer Science in Cyber Security (07BCCY)
Nature of Case: **(Lab Objective)**	The purpose of this lab is to demonstrate the ability to perform mobile forensic evidence extraction and analysis using forensic tools and techniques. The lab focuses on acquiring data from Apple mobile devices using trusted computer authentication methods and logical acquisition through forensic software. The investigation also demonstrates how lockdown files can be used to bypass trust verification on iOS devices and how Belkasoft Evidence Center can perform logical acquisition of device data for forensic analysis.
Due Date:	*16 March 2026*

Description of Evidence		**Vendor Name**	**Model /Serial No.**
Item #1	Apple iPhone device	Apple Inc	iPhone 15 pro max
Item #2	Lockdown Files (.plist files)	Apple	USB-C
Item #3	Belkasoft Acquisition Tool	Belkasoft	Software Tool
Item #4	libimobiledevice	Open Source	Software Tool v1.3.0
Item #5	Forensic Workstation Windows 10	HP	HP 250 15.6- inch G9 notebook PC (7N0M6ES)

1. Introduction

Mobile devices contain large amounts of digital evidence that can be used in forensic investigations. Mobile forensics involves the extraction, preservation, and analysis of data from mobile devices in a forensically sound manner. iOS devices implement strong security mechanisms such as passcodes and trusted computer verification to prevent unauthorized access.

However, once an iOS device trusts a computer, a pairing record known as a lockdown file is created. These files allow the trusted computer to communicate with the device without requiring the user to unlock it again. In forensic investigations, analysts can use these pairing records to access device data. Tools such as Belkasoft Evidence Center enable investigators to perform logical acquisitions such as iTunes backups to extract information including contacts, messages, and application data for analysis.

2. Lab Procedures

2.1 Task 1 – Password Protection and Potential Bypasses Using Lockdown Files

The objective of this task was to demonstrate how lockdown files stored on trusted computers can allow a forensic workstation to communicate with an iOS device without requiring user interaction.

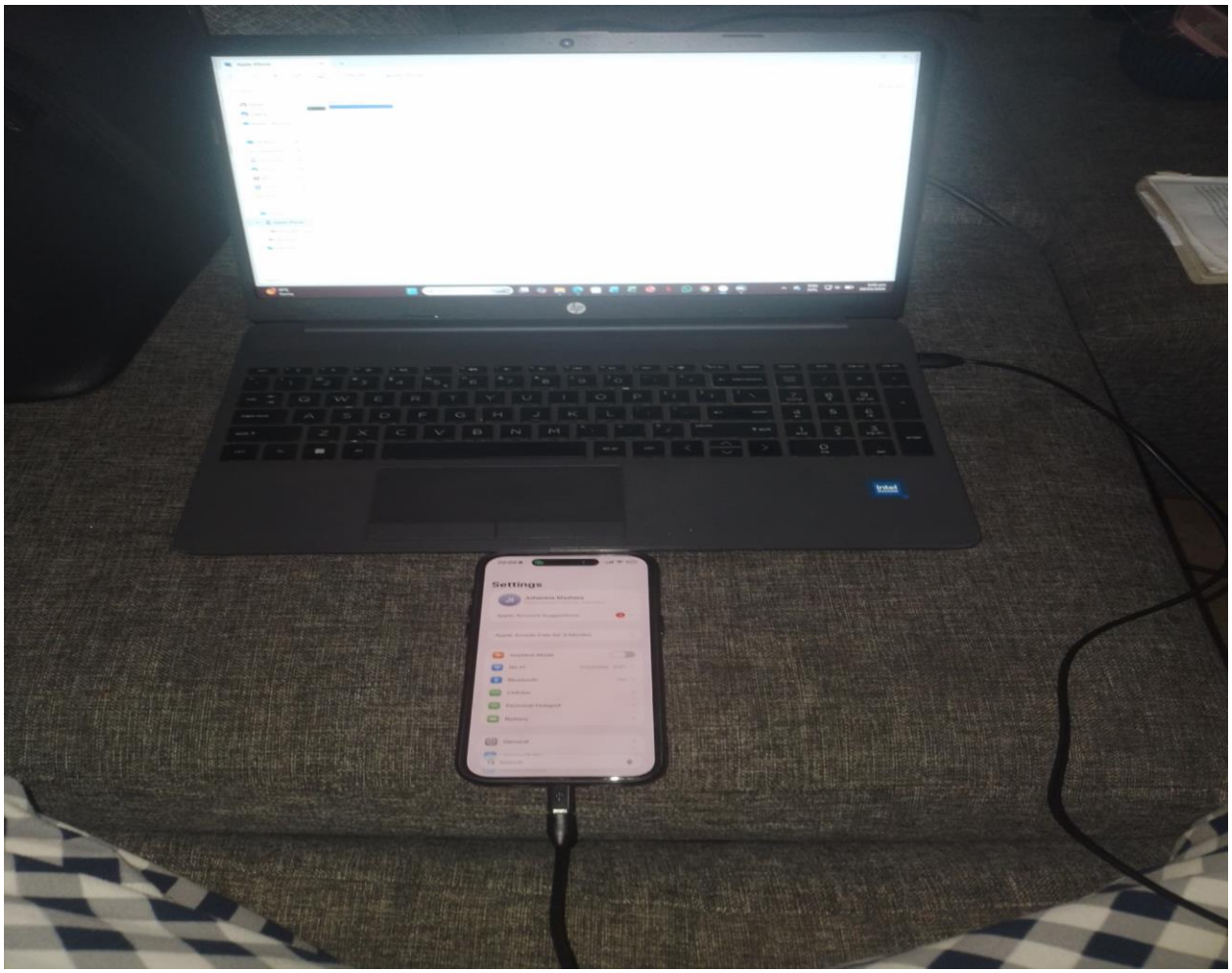
Step 1: Unlock the Device

The iPhone device was first unlocked using the correct passcode.

This step is important because iOS devices only create or update pairing records when the device has been unlocked at least once after a reboot.

Screenshot 1:

Insert screenshot showing the unlocked iPhone device.



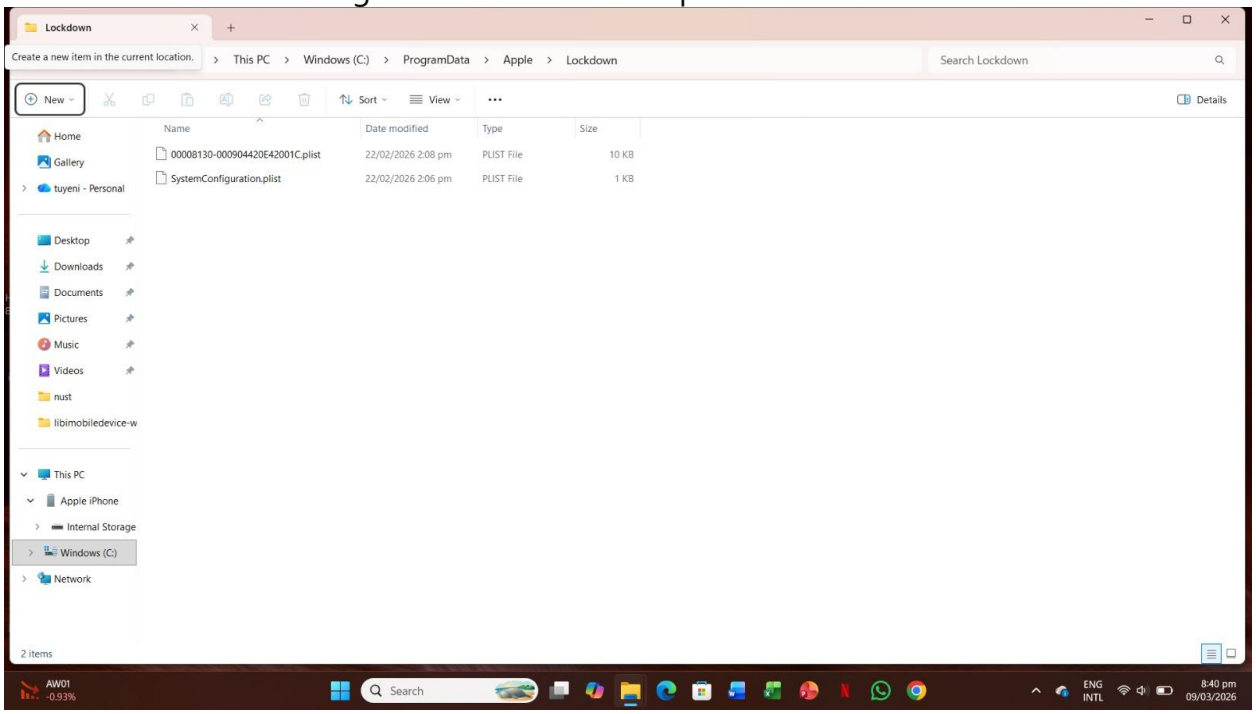
Step 2: Connect iPhone to Trusted Computer
The iPhone device was connected to the forensic workstation using a USB cable.
The device prompted a message asking "Trust This Computer?".
The Trust option was selected.
📸 **Screenshot 2:**
Insert screenshot showing the Trust This Computer prompt.

Step 3: Locate Lockdown Files
Lockdown files were located on the forensic workstation in the following directory:
Windows location:
C:\ProgramData\Apple\Lockdown

These files are stored as .plist files and contain pairing records used to authenticate t rusted computers.

📸 **Screenshot 3:**

Insert screenshot showing the Lockdown folder path and files.



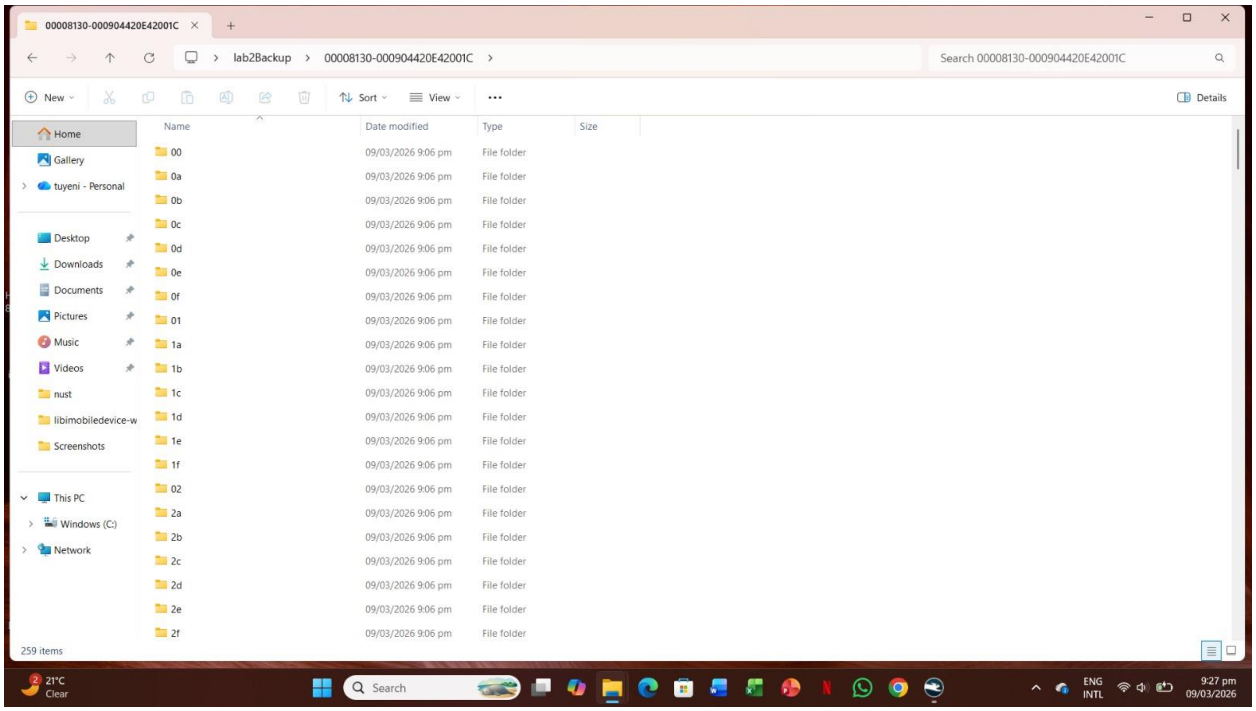
Step 4: Copy Lockdown Files

The lockdown files were copied from the trusted computer and transferred to the for ensic workstation.

This step allows the forensic workstation to use the pairing record to communicate w ith the device as if it were the original trusted computer.

📸 **Screenshot 4:**

Insert screenshot showing lockdown files being copied.



Step 5: Place Lockdown Files on Forensic Workstation

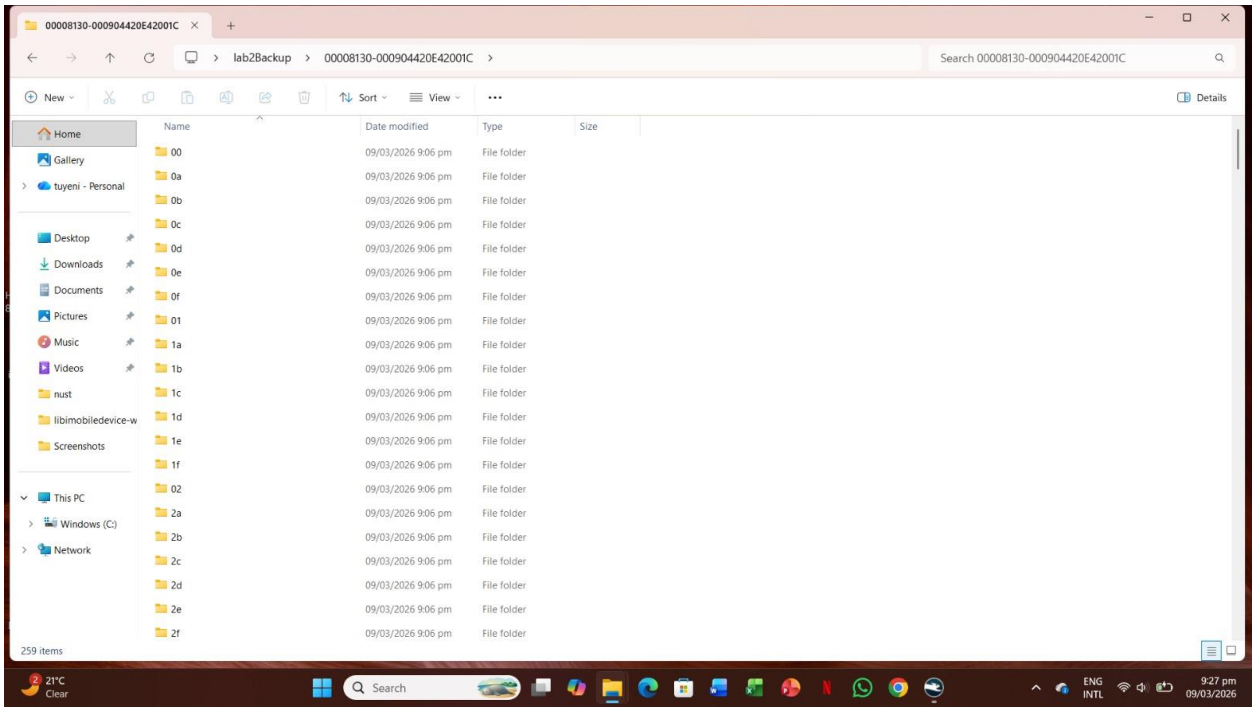
The copied lockdown files were placed into the following directory on the forensic w orkstation:

C:\ProgramData\Apple\Lockdown

This tricks the device into believing the forensic workstation is the previously trusted computer.

📸 **Screenshot 5:**

Insert screenshot showing lockdown files placed in destination folder.



Step 6: Verify Device Connection Using libimobiledevice

The libimobiledevice tool was used to verify the connection between the device and the workstation.

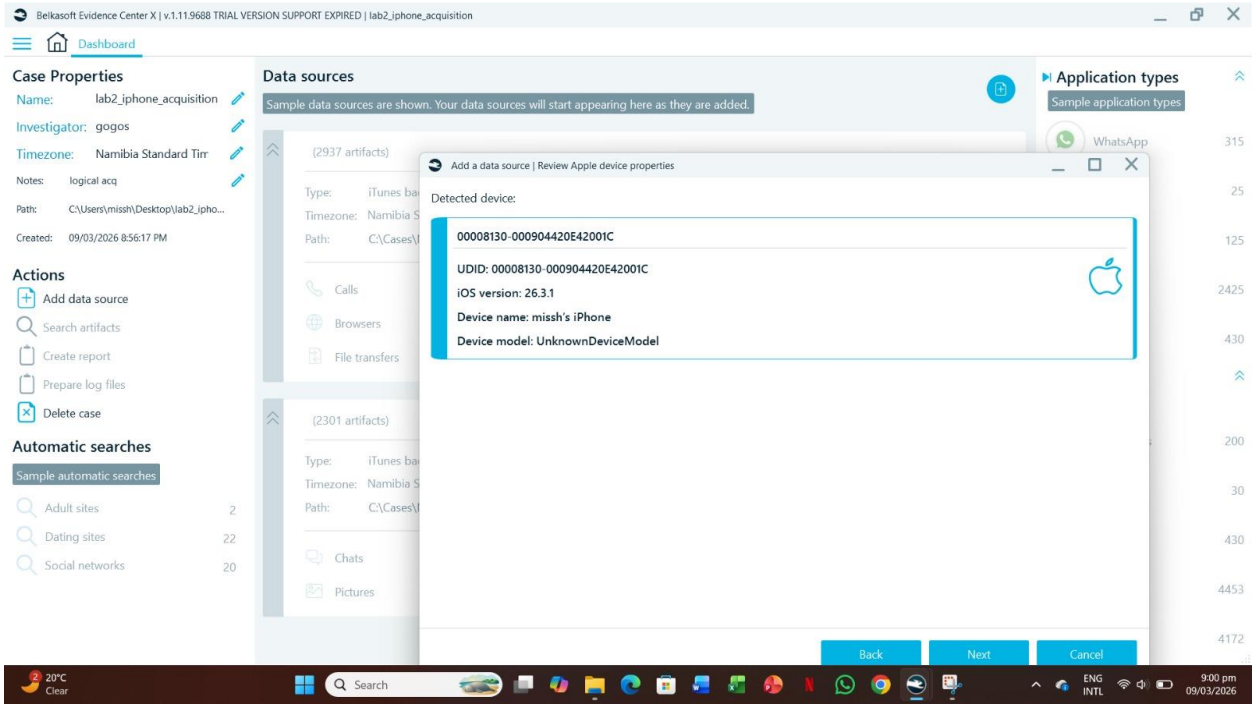
Command used:

idevice_id -l

This command lists connected iOS devices.

📸 **Screenshot 6:**

Insert screenshot showing device detected using terminal command.



2.2 Task 2 – Logical Acquisition with Belkasoft

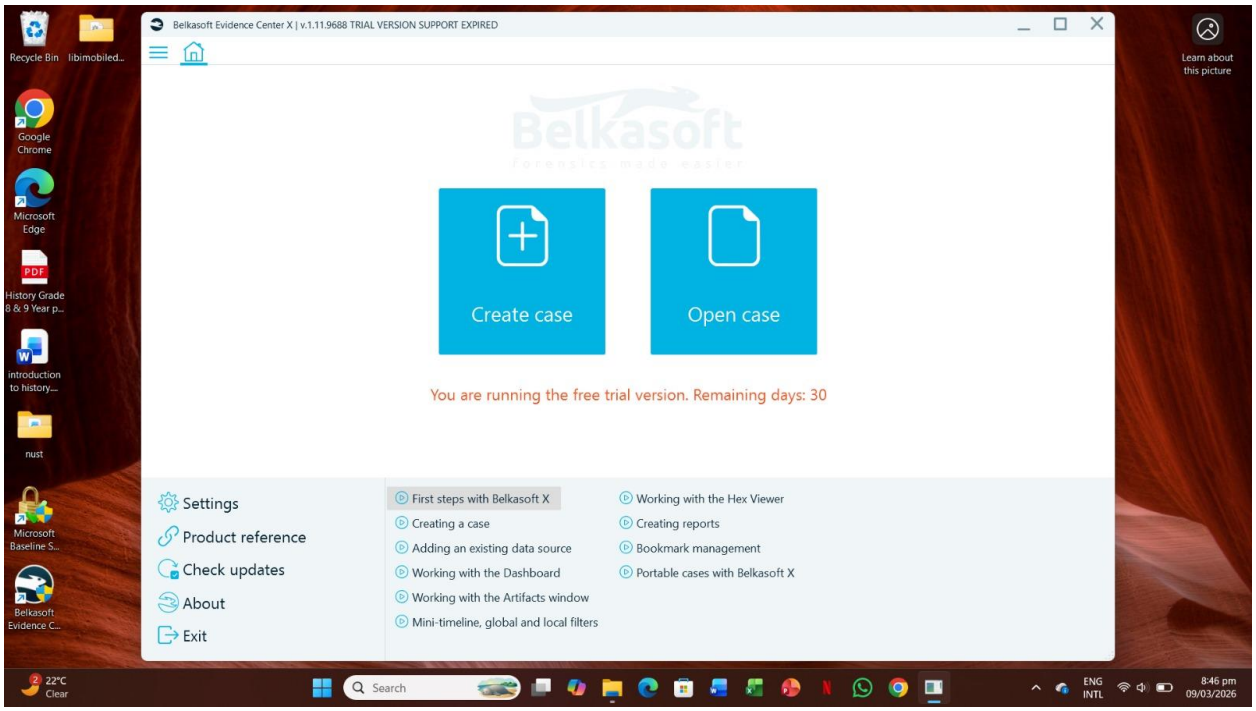
This task involved acquiring a logical image of the iPhone device using Belkasoft Acq uisition Tool.

Step 1: Launch Belkasoft Acquisition Tool

The Belkasoft Acquisition Tool was opened on the forensic workstation.

📸 **Screenshot 7:**

Insert screenshot showing Belkasoft tool opening screen.

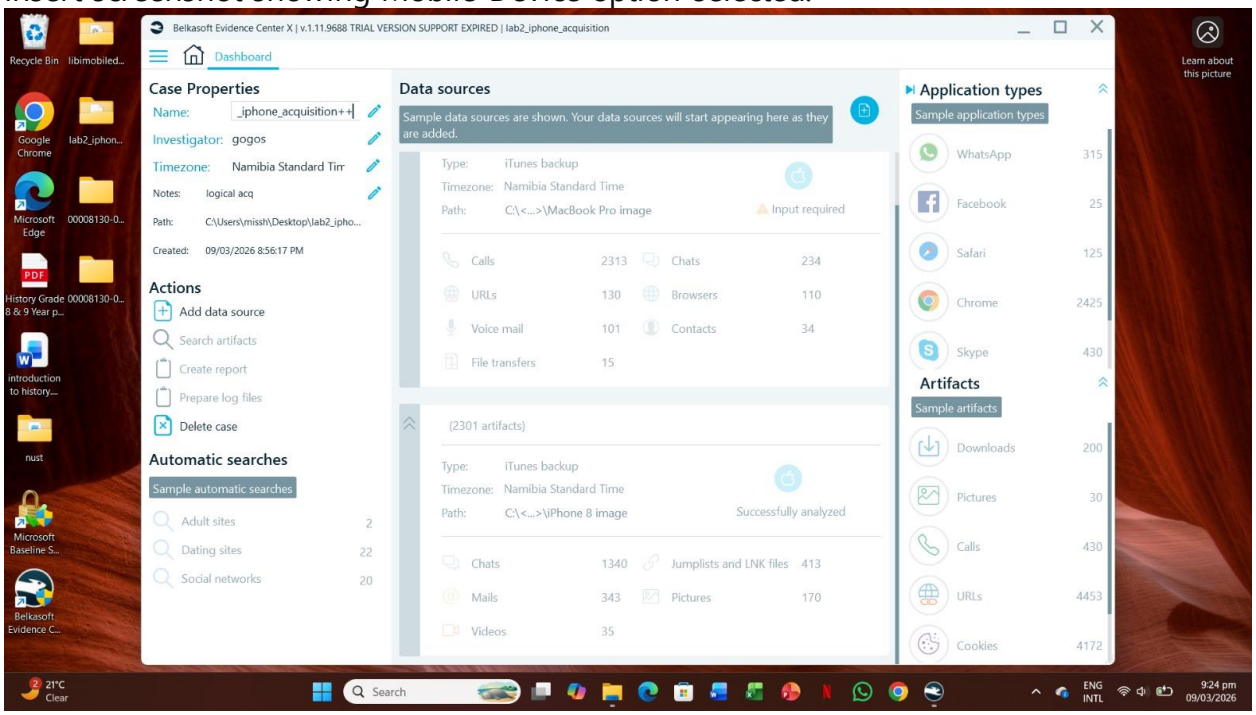


Step 2: Select Mobile Device Option

From the acquisition options menu, the Mobile Device option was selected.

📸 **Screenshot 8:**

Insert screenshot showing Mobile Device option selected.

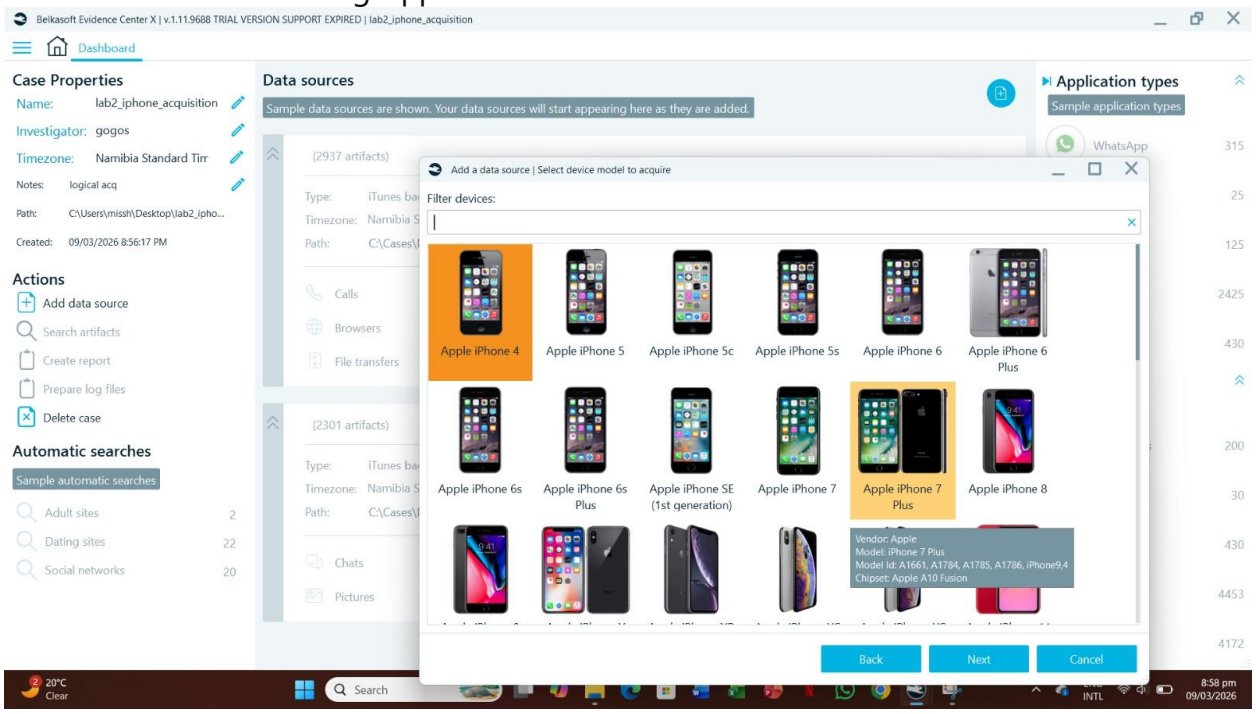


Step 3: Select Apple Device

In the next window, the Apple option was selected to indicate that the acquisition wil l be performed on an iOS device.

📸 **Screenshot 9:**

Insert screenshot showing Apple device selection.



Step 4: Choose Acquisition Method

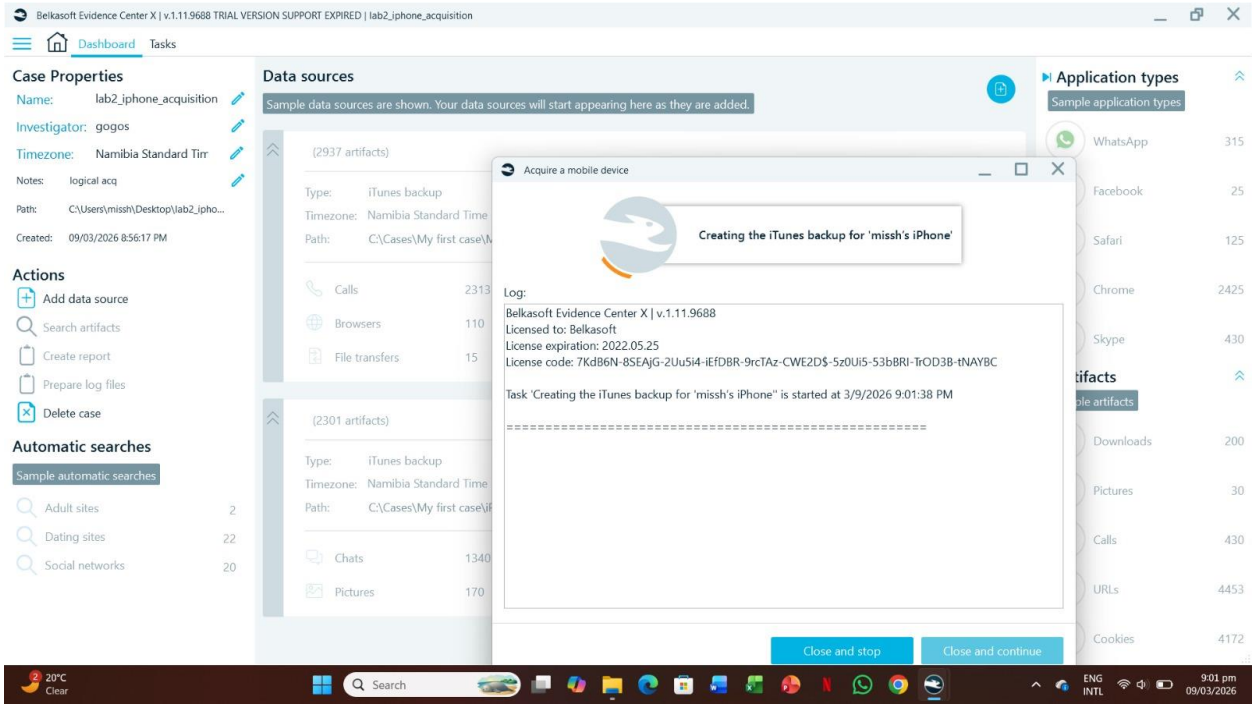
The acquisition method selected was:

iTunes Backup

This method allows logical acquisition of the device without requiring the device to b e jailbroken.

📸 **Screenshot 10:**

Insert screenshot showing iTunes Backup option selected.



Step 5: Select Image Path

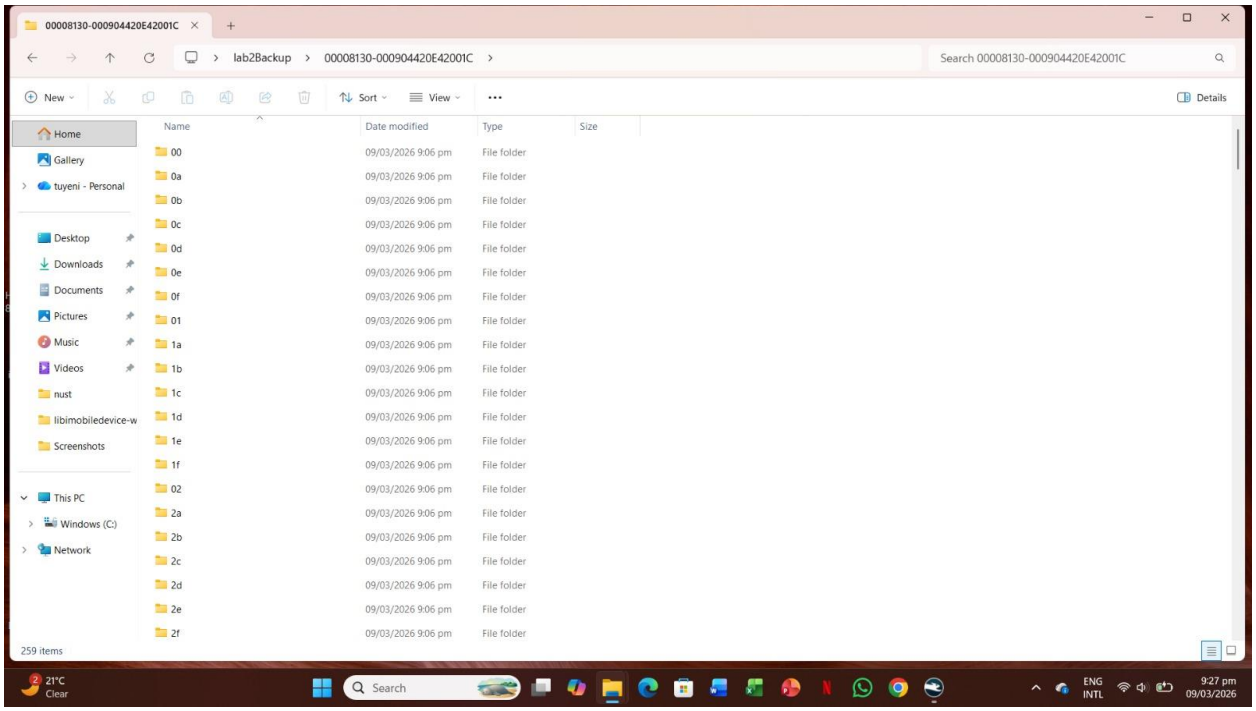
The destination folder for the backup image was selected.

Example path:

C:\MobileForensics\iPhoneBackup\

📸 **Screenshot 11:**

Insert screenshot showing backup destination folder selection.



Step 6: Start Acquisition Process

The acquisition process was started and the tool began creating a logical backup of t he device.

During this stage, the tool extracted files such as:

Contacts

Messages

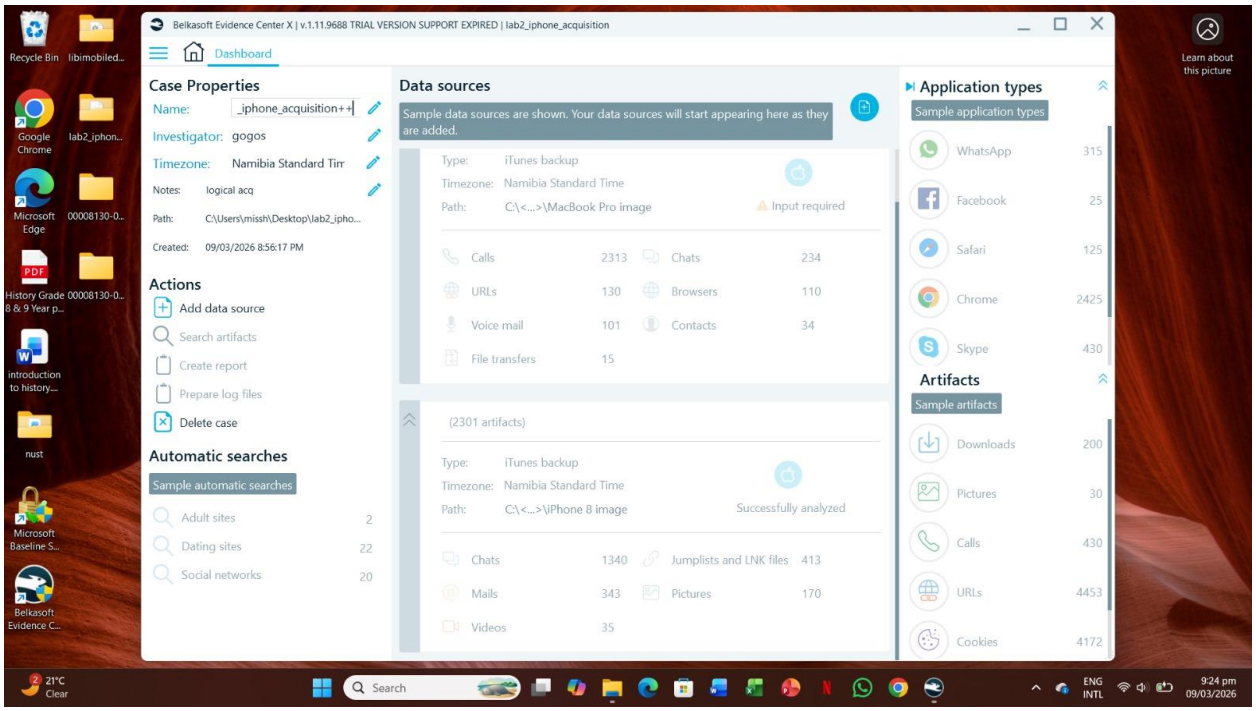
Photos

Application data

System information

📸 **Screenshot 12:**

Insert screenshot showing acquisition process running.



Step 7: Acquisition Completed

Once the process was completed successfully, the logical backup files were saved in t he selected destination folder.

📸 **Screenshot 13:**

Insert screenshot showing successful acquisition message.

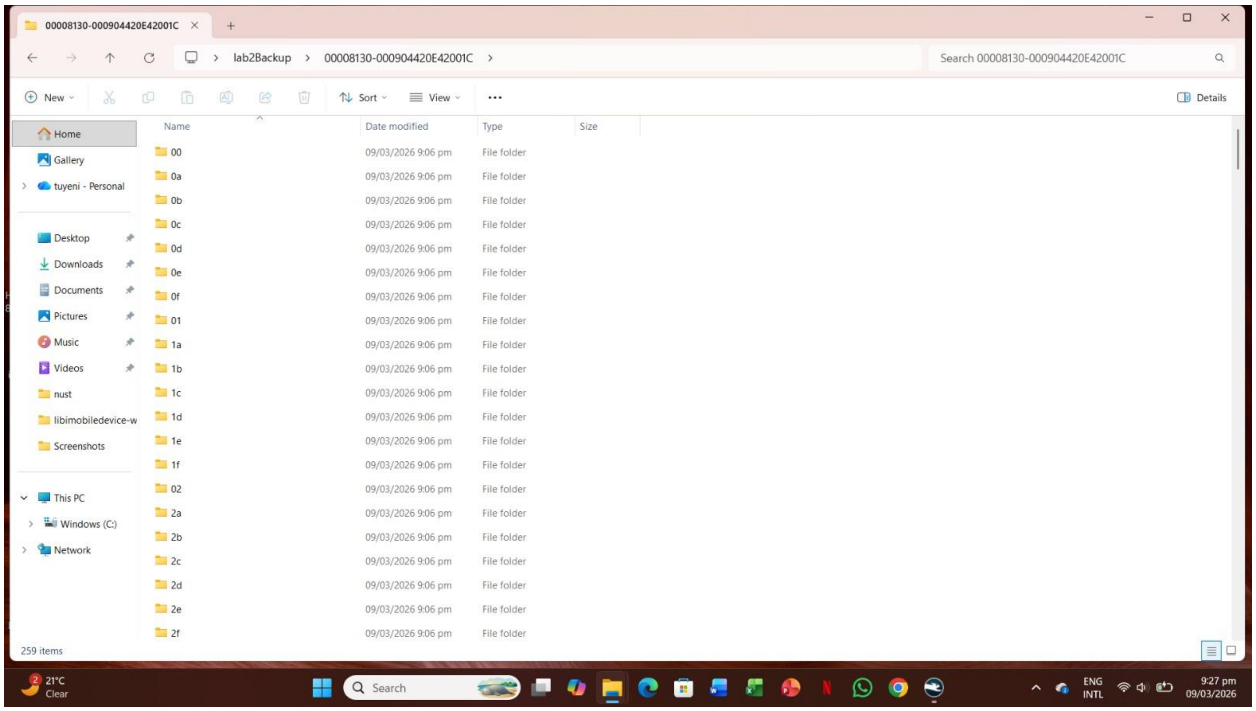
Step 8: Verify Logical Image

The created backup was verified in the destination directory.

The backup can later be analysed using Belkasoft Evidence Center or other forensic t ools.

📸 **Screenshot 14:**

Insert screenshot showing backup files in folder.



3. Lab Challenges/Problems Encountered

During the lab investigation, several challenges were encountered:

1. Locating the ProgramData folder required enabling hidden files in Windows since it is hidden by default.
2. The iPhone device initially failed to appear in the Belkasoft tool until the correct A pple drivers were installed.
3. The device had to be unlocked at least once after reboot before the pairing recor ds could be used.

Despite these challenges, the investigation was successfully completed.

4. Conclusion

This lab demonstrated how forensic investigators can access iOS devices using tr usted computer pairing records known as lockdown files. By copying these files t o a forensic workstation, investigators can bypass the need for user interaction an d establish communication with the device.
Additionally, the lab demonstrated the use of Belkasoft Acquisition Tool to perfor m logical acquisition through an iTunes backup. Logical acquisition provides inve stigators with valuable data such as contacts, messages, and application files. The se techniques are important in digital forensic investigations where mobile device s are involved.

5. The role played by each group member:

Barnabas Iihuhwa 223081701 - Performed device preparation, unlocked the device, and captured screenshots for Task 1.

Tjijenda.u 220113009 – Managed evidence collection

Titus Shaangelo 22084774 -Documented the procedures, compiled the report, and organised screenshots.

References

Belkasoft.
(2023). *Belkasoft Evidence Center User Guide*. Available at: https://belkasoft.com

Satvat, K., & Hubballi, N.
(2020). *Practical Mobile Forensics: Investigate, Analyze and Secure Mobile Device Data (4th Edition)*. Packt Publishing.

Apple Inc. (2023). *iOS Security Guide*. Available at: https://support.apple.com